



DP-600 : Implement analytics solutions using Microsoft Fabric

Exercise 10 : Design a semantic model for scale

- **Create a workspace**
- **Create a Lakehouse**
- **Create a semantic model**
- **Design star schema relationships**
- **Create measures**
- **Create a calculation group**
- **Use USERELATIONSHIP for the inactive ship date relationship**
- **Validate the model with a report**



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Create a workspace

Home

Workspaces

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

semantic_model_scale_10

My workspace

...

semantic_model_scale_10

Create deployment pipeline

Create app

Manage access

Workspace settings

+ New item

New folder

Import

Migrate

Filter by keyword

Filter

Choose from predefined task flows or add a task to build one

Select from one of Microsoft's predefined task flows or add a task to start building one yourself.

Select a predefined task flow

Add a task

Import a task flow

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
Create-Sales-Data		Notebook	—	Sidv...	—	—	—	—	
SalesLakehouse		Lakehouse	—	Sidv...	—	—	—	—	
SalesLakehouse		SQL analytic...	—	S...	—	—	—	—	

WS_Name: semantic_model_scale_10

WS_Type: Fabric Trial



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➤ Create a Lakehouse

New Lakehouse

Name *

Location *

 semantic_model_scale_10 ▼



☒ Lakehouse schemas ⓘ

CreateCancel



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➤ Create a semantic model

New semantic model

Direct Lake semantic model name ⓘ
Sales Model

Workspace ⓘ
semantic_model_scale_10

Storage mode
☐ Direct Lake on OneLake ⓘ ☒ Direct Lake on SQL ⓘ

Select or deselect tables for the semantic model.
Search  

☒ Select all

☒  dbo

☒  dimcustomer

☒  dimdate

☒  dimproduct

☒  factsales

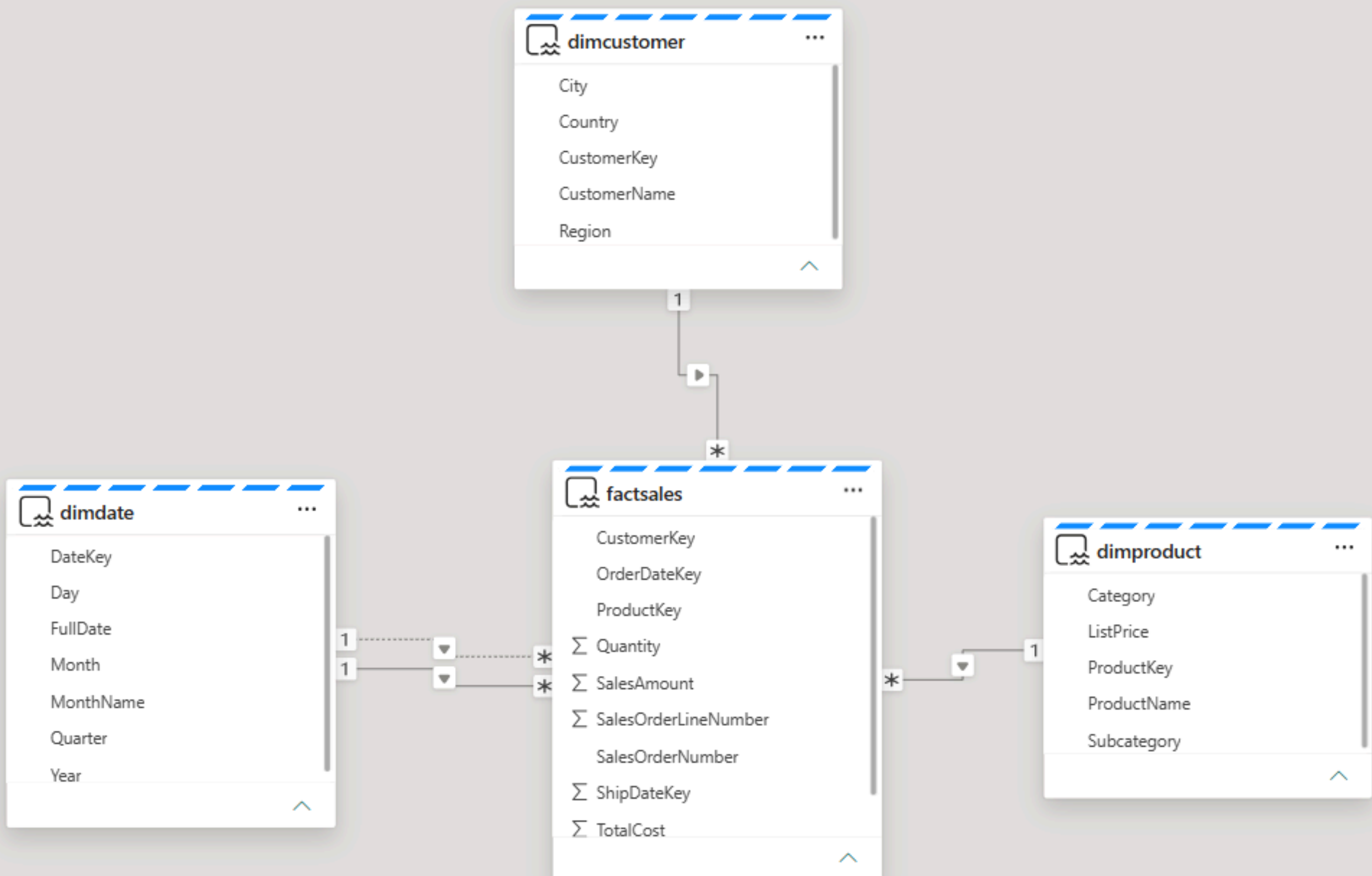
[Learn more about Direct Lake](#) 

Confirm Cancel



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➤ Design star schema relationships





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➤ Create measures

« **Data** »

Properties

Tables Model

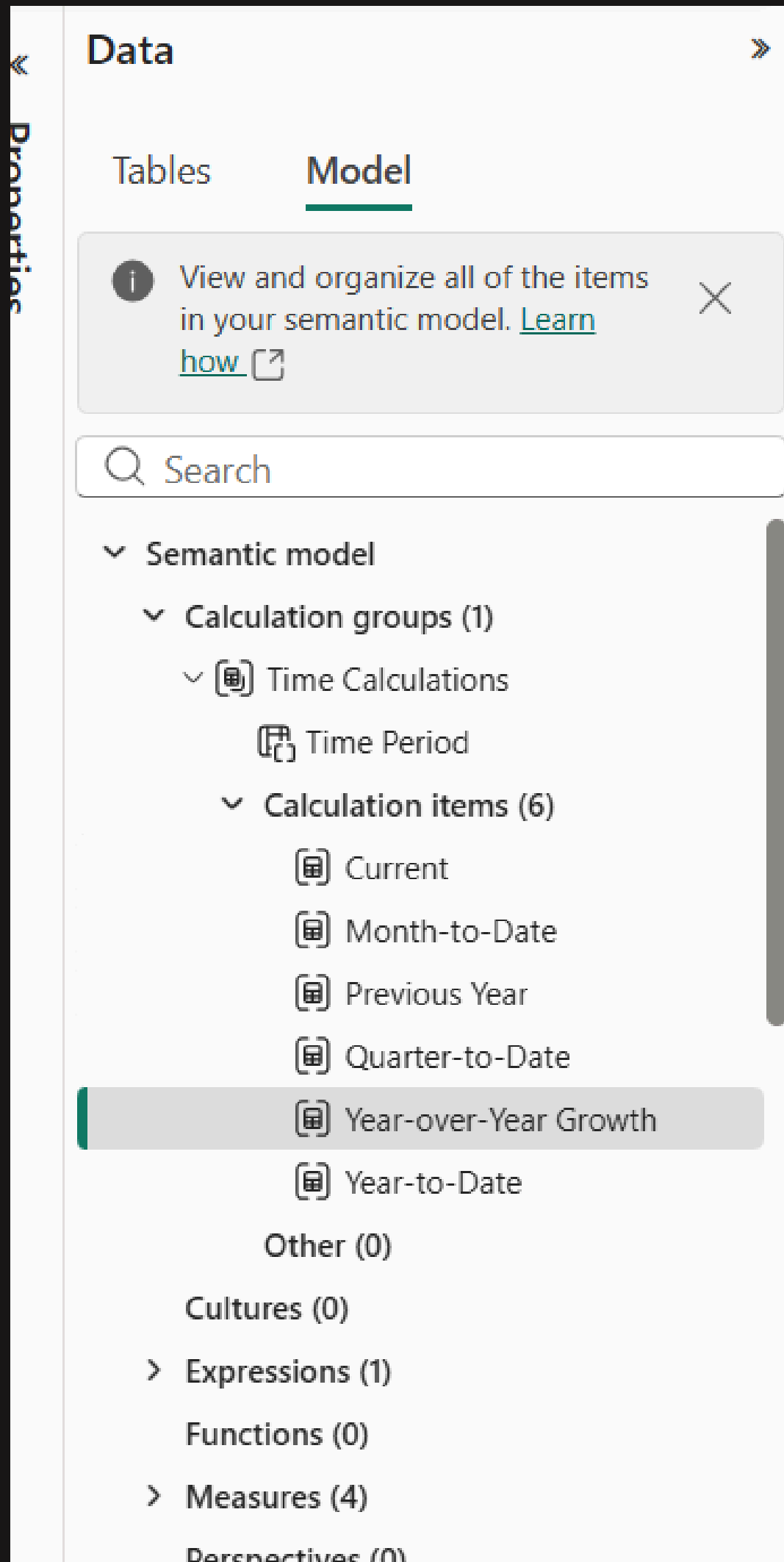
Search

- > dimcustomer
- > dimdate
- > dimproduct
- ▼ factsales
 - CustomerKey
 - OrderDateKey
 - ProductKey
 - Profit
 - Profit Margin
 - Σ Quantity
 - Σ SalesAmount
 - Σ SalesOrderLineNumber
 - SalesOrderNumber
 - Σ ShipDateKey
 - Total Cost
 - Total Sales
 - Σ TotalCost
 - Σ UnitPrice



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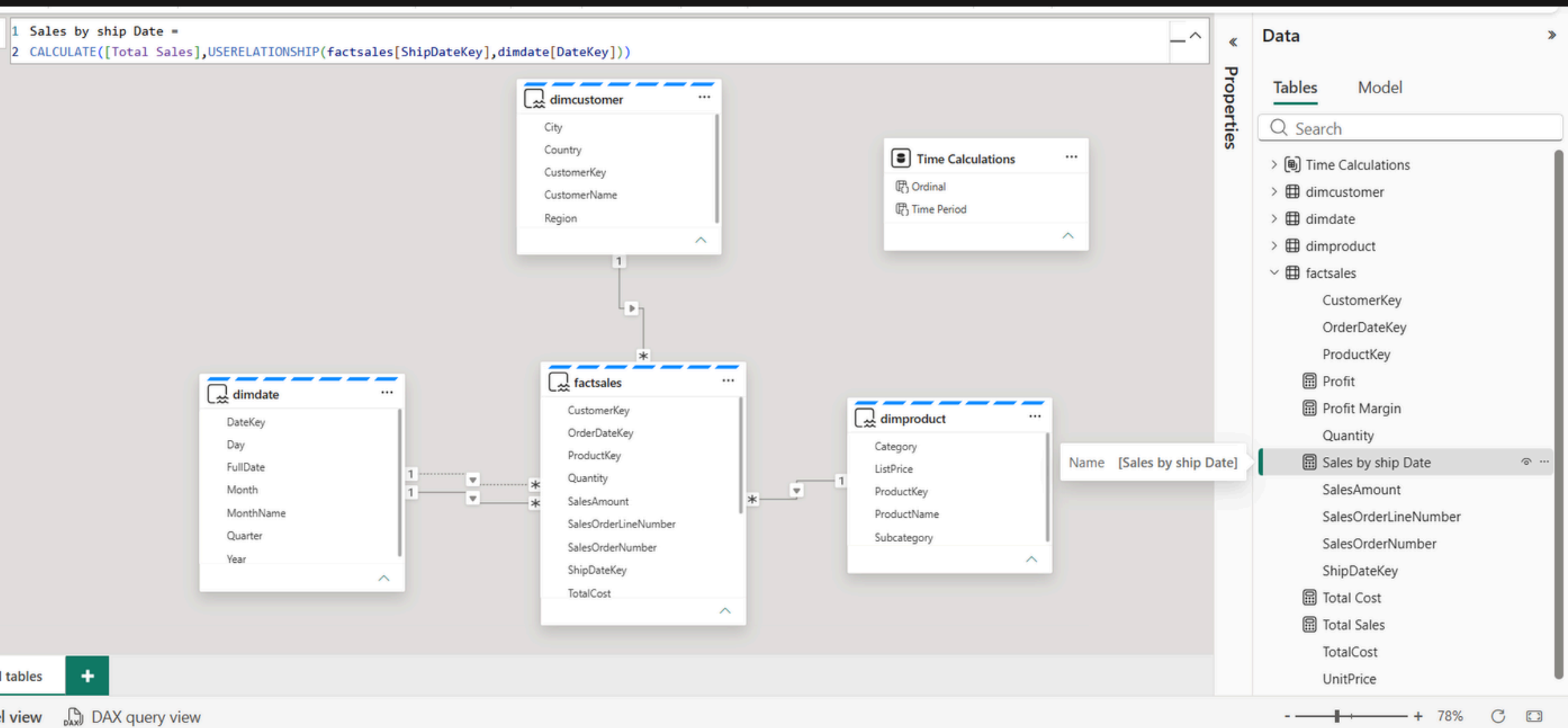
➤ Create a calculation group





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- Use USERELATIONSHIP for the inactive ship date relationship



**Sales by Ship Date =
CALCULATE(
[Total Sales],
USERELATIONSHIP(FactSales[ShipDateKey], DimDate[DateKey])
)**



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Configure settings for scale

Query scale-out

Isolate refresh operations from query processing and automatically distribute queries across multiple replicas when query volume is high.

☒ On

Apply

Discard

OneLake Integration

You can automatically write data imported into your semantic model tables to delta tables in OneLake. Make sure your semantic model includes one or more import tables. [Learn more](#)

☒ On

Apply

Discard

Semantic model Image

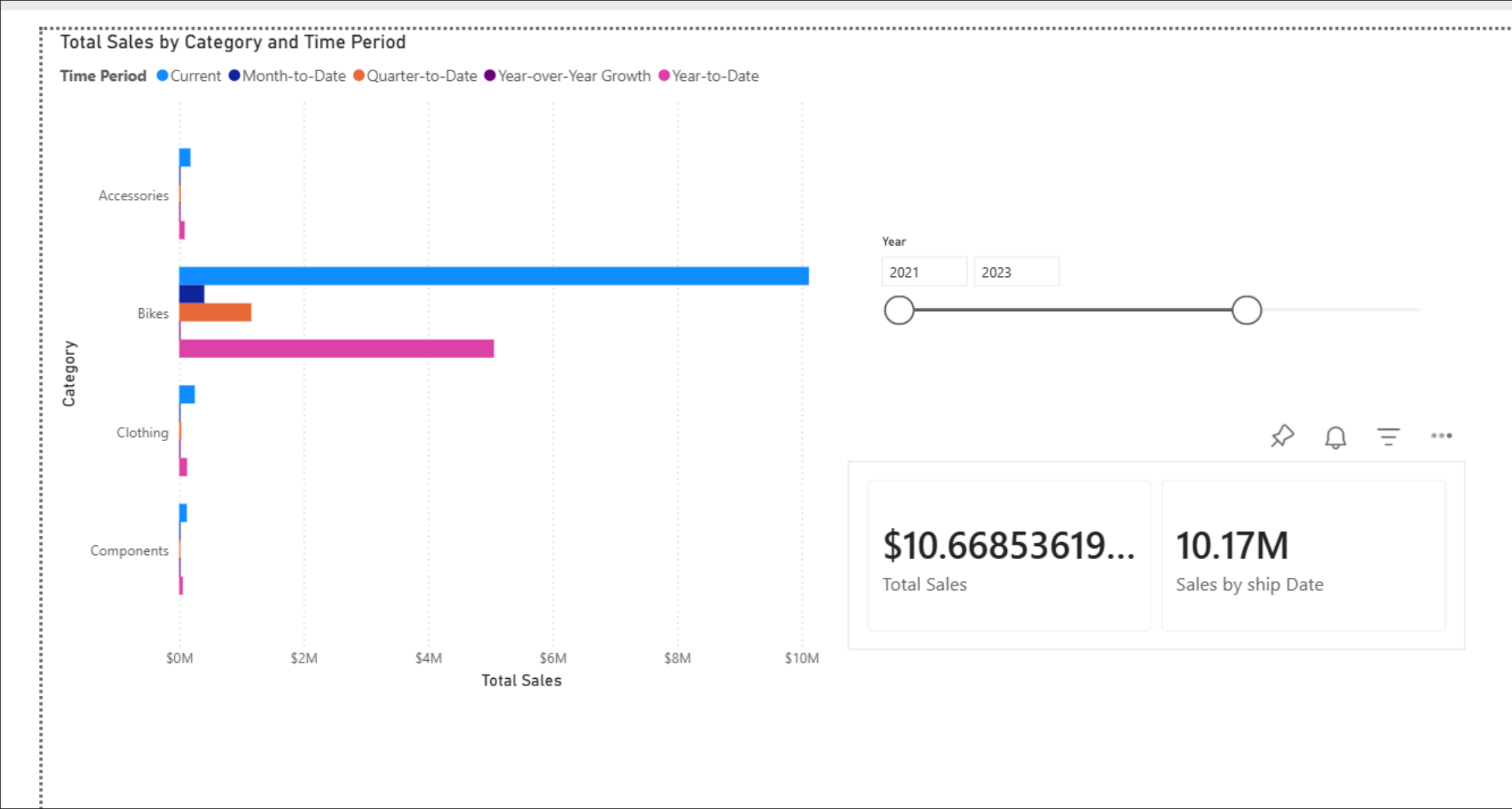
External sharing



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Validate the model with a report





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Reference

